

UNIVERSITY EXAMINATION

CELL BIOLOGY & GENETICS

Time Allowed: 1 Hour

Maximum Marks: 20

GENERAL INSTRUCTIONS:

- 1. Answer all questions according to section guidelines.
- 2. Figures to the right indicate full marks allocated to each question.

SECTION A - MULTIPLE CHOICE QUESTIONS

Select the correct alternative for each question.

Q1. Which scientists proposed the Fluid Mosaic Model of the cell membrane structure in 1972? [1]

- | | |
|---|------------------------------------|
| A. Matthias Schleiden and Theodor Schwann | B. S. J. Singer and G. L. Nicolson |
| C. Hugh Davson and James Danielli | D. James Watson and Francis Crick |

Q2. In a Mendelian monohybrid cross between two heterozygous pea plants ($Tt \times Tt$), what is the expected genotypic ratio among the offspring? [1]

- | | |
|--------|------------|
| A. 3:1 | B. 1:2:1 |
| C. 1:1 | D. 9:3:3:1 |

Q3. What primary function does DNA helicase perform at the replication fork during semi-conservative DNA replication? [1]

- | | |
|--|---|
| A. Synthesizes short RNA primers on the template strand | B. Relieves torsional strain created by supercoiling |
| C. Unwinds double-stranded DNA by breaking hydrogen bonds between base pairs | D. Covalently joins Okazaki fragments on the lagging strand |

Q4. How does cholesterol influence membrane fluidity at high physiological temperatures? [1]

- | | |
|--|---|
| A. Increases fluidity by separating phospholipid tails | B. Restrains movement of phospholipids, thereby decreasing fluidity |
| C. Disrupts integral protein conformation to make the membrane leaky | D. Enhances lateral diffusion of all membrane lipids |

Q5. In guinea pigs, black coat color (B) is dominant over white coat color (b). A test cross of a black guinea pig yields 50% black and 50% white offspring. What is the genotype of the black parent? [1]

- | | |
|-------|--|
| A. BB | B. Bb |
| C. bb | D. Undetermined without F2 phenotypic data |

SECTION B - SHORT ANSWER QUESTIONS

Answer all questions concisely giving scientific explanations and diagrams where appropriate.

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- Q6.** Describe the Fluid Mosaic Model of cell membranes proposed by Singer and Nicolson. Explain the structural basis for both the 'fluid' and 'mosaic' nature of the membrane. [5]
- Q7.** A dihybrid pea plant heterozygous for seed color and seed shape (YyRr - Yellow, Round) is crossed with a double recessive plant (yyrr - Green, Wrinkled). a) List all possible gametes produced by each parent. b) Construct a Punnett square for this test cross. c) Provide the predicted phenotypic ratio of the offspring. [5]
- Q8.** Contrast leading strand and lagging strand synthesis during semi-conservative DNA replication. Explain why continuous synthesis occurs on one strand while discontinuous synthesis occurs on the other, identifying all major enzymes involved in lagging strand completion. [5]

CONFIDENTIAL - OFFICIAL EVALUATION SCHEME & ANSWER KEY

SECTION A - Multiple Choice Questions - Solutions

Q1. (MCQ, 1M) [Topic: Unit 1: Cell Membrane Structure & Fluid Mosaic Model | Level: Remember]

Answer/Rubric: B. S. J. Singer and G. L. Nicolson. In 1972, S. J. Singer and Garth L. Nicolson proposed the Fluid Mosaic Model to explain the structure and dynamics of biological membranes.

Q2. (MCQ, 1M) [Topic: Unit 2: Mendelian Genetics & Punnett Squares | Level: Remember]

Answer/Rubric: B. 1:2:1 - A monohybrid cross of two heterozygotes yields offspring with a genotypic ratio of 1 TT : 2 Tt : 1 tt.

Q3. (MCQ, 1M) [Topic: Unit 3: DNA Replication & Mitosis | Level: Understand]

Answer/Rubric: C. Unwinds double-stranded DNA by breaking hydrogen bonds between base pairs - DNA helicase uses ATP energy to break hydrogen bonds between complementary base pairs, separating the two template strands.

Q4. (MCQ, 1M) [Topic: Unit 1: Cell Membrane Structure & Fluid Mosaic Model | Level: Analyze]

Answer/Rubric: B. Restrains movement of phospholipids, thereby decreasing fluidity - Cholesterol acts as a temperature buffer; at high temperatures, it stiffens the membrane by restricting the excessive lateral movement of phospholipid fatty acid chains.

Q5. (MCQ, 1M) [Topic: Unit 2: Mendelian Genetics & Punnett Squares | Level: Apply]

Answer/Rubric: B. Bb - A test cross crosses a phenotypically dominant individual with a homozygous recessive individual (bb). Producing a 1:1 phenotypic ratio confirms the parent was heterozygous (Bb).

SECTION B - Short Answer Questions - Solutions

Q6. (Short Answer, 5M) [Topic: Unit 1: Cell Membrane Structure & Fluid Mosaic Model | Level: Understand]

Answer/Rubric: 1. Overview: Proposed in 1972, the model describes the plasma membrane as a dynamic, two-dimensional fluid matrix composed of a phospholipid bilayer with embedded and attached proteins. 2. Structural Basis of 'Fluid': The phospholipid bilayer is held together by weak hydrophobic interactions, allowing lipids and many proteins to diffuse laterally within the plane of the membrane. 3. Structural Basis of 'Mosaic': The membrane features a mosaic pattern created by various functional biomolecules, including integral membrane proteins, peripheral membrane proteins, glycoproteins, glycolipids, and cholesterol intercalated throughout the bilayer. 4. Importance: Fluidity ensures proper membrane transport, signaling, cell division, and cell-to-cell recognition.

Q7. (Short Answer, 5M) [Topic: Unit 2: Mendelian Genetics & Punnett Squares | Level: Apply]

Answer/Rubric: a) Gametes: - YyRr parent produces: YR, Yr, yR, yr - yyrr parent produces: yr b) Punnett Square: | YR | Yr | yR | yr - yr | YyRr | Yyrr | yyRr | yyrr c) Phenotypic Ratio: - 1 Yellow, Round (YyRr) : 1 Yellow, Wrinkled (Yyrr) : 1 Green, Round (yyRr) : 1 Green, Wrinkled (yyrr) (Ratio 1:1:1:1).

Q8. (Short Answer, 5M) [Topic: Unit 3: DNA Replication & Mitosis | Level: Analyze]

Answer/Rubric: 1. Cause of Discontinuous Synthesis: DNA polymerase can only add nucleotides in the 5' to 3' direction. Because template DNA strands are antiparallel (5'→3' and 3'→5'), as the replication fork opens, only one template strand allows continuous synthesis toward the fork (leading strand). 2. Leading vs. Lagging Strand: - Leading Strand: Synthesized continuously toward the replication fork using a single RNA primer. - Lagging Strand: Synthesized discontinuously away from the replication fork in short segments called Okazaki fragments. 3. Enzymes in Lagging Strand Completion: - Primase: Synthesizes multiple short RNA primers to initiate each Okazaki fragment. - DNA Polymerase III: Extends fragments from 5' to 3' until reaching the adjacent primer. - DNA Polymerase I: Removes RNA primers via exonuclease activity and replaces them with DNA nucleotides. - DNA Ligase: Forms phosphodiester bonds to join adjacent Okazaki fragments into a continuous strand.